



Smart Attendance System

A Web-Based Attendance & Marks Management Platform

PHP

MySQL

HTML

CSS

XAMPP

Project Overview

What is it?

A web-based system to automate attendance and marks management in educational institutions.

Eliminates paper-based processes and reduces human errors.

Provides separate portals for Admin and Students.



6

Modules



4

DB Tables



2

User Roles



6+

Tech Stack

Problem Statement & Proposed Solution

PROBLEMS

Manual Attendance Issues

- Attendance sheets can be lost or damaged
- Calculating attendance percentage manually is difficult
- Managing student marks manually is time-consuming
- Human errors occur frequently in record keeping
- Students cannot easily view their own marks

SOLUTION

Smart Digital Platform

- Digital records stored securely in MySQL database
- Automatic attendance percentage calculation
- Admin can add, edit, and delete marks with ease
- Role-based authentication prevents data tampering
- Student portal for viewing personal marks & profile

System Features



Admin Features

- Admin Login & Authentication
- Add / Manage Students
- Take & Delete Attendance
- Add / Delete Marks
- Attendance & Marks Reports
- Percentage Report Generation



Student Features

- Student Registration
- Secure Student Login
- View Personal Profile
- View Personal Marks
- View Course Results
- Read-Only Access (Secure)

Database Design

attendance_db

users

Stores login info for admins & students

id – INT (PK)
username – VARCHAR
password – VARCHAR
role – VARCHAR
student_id – VARCHAR

attendance

Records daily attendance status

id – INT (PK)
student_id – VARCHAR (FK)
status – VARCHAR
date – DATE

students

Stores student personal details

student_id – VARCHAR (PK)
name – VARCHAR
department – VARCHAR

marks

Stores exam scores per course

id – INT (PK)
student_id – VARCHAR (FK)
course_name – VARCHAR
mid_marks – INT
final_marks – INT

Technology Stack

PHP

Backend Logic & Server-side Processing

MySQL

Relational Database Storage

HTML

Web Page Structure & Markup

CSS

Styling & Responsive Design

JS

Interactive UI Features

XAMPP

Local Development Server

System Modules & Workflow

01 Authentication

Role-based login for Admin & Students using PHP sessions

02 Attendance

Mark Present/Absent, save records, delete entries

03 Student Mgmt

Add students, manage profiles and departments

04 Marks Mgmt

Insert mid/final marks, auto-grade calculation

05 Reports

Attendance %, marks summary, and grade reports

06 Student Portal

Students view their own marks and profile (read-only)

Marks & Grading System

	Total Marks	Grade	Performance
	80 – 100	 A+ Excellent	
	70 – 79	 A Very Good	
	60 – 69	 B Good	
	50 – 59	 C Satisfactory	
	Below 50	 F Fail	

Total = Mid Marks + Final Marks

Advantages & Limitations

✓ ADVANTAGES

- ✓ Saves time with automated attendance
- ✓ Reduces manual paperwork significantly
- ✓ Easy and fast report generation
- ✓ Better data security and accuracy
- ✓ Modern, user-friendly interface
- ✓ Faster attendance percentage calculation
- ✓ Secure role-based access control

✗ LIMITATIONS

- ✗ Works on local server only (XAMPP)
- ✗ No email or SMS notification system
- ✗ No dedicated mobile application
- ✗ No biometric attendance integration

Future Improvements



Mobile App

Native Android/iOS application for students and teachers



Cloud Hosting

Deploy on cloud (AWS/GCP) for remote access



Email & SMS

Automated notifications for low attendance alerts



Biometric

Fingerprint or card-based attendance scanning



Analytics

Data analytics dashboard with visual attendance trends



Face Recognition

AI-powered automated face-based attendance marking



Conclusion

The Smart Attendance System successfully automates attendance and marks management using PHP and MySQL. The project reduces manual work, improves accuracy, and provides a secure platform for both administrators and students.

- Practical CRUD operations on a real database
- Role-based authentication with PHP sessions
- Modern UI with sidebar navigation & responsive tables
- Scalable architecture for future enhancements

Thank You